1. Understand and utilise constructs based on definitions, rules and formal systems including employing familiar algorithms (Point 5 in the list)

This is the core ability required to solve this problem. Students must understand and apply mathematical rules, especially the algebraic rules regarding inverse proportional relationships between variables.

- **Specific Analysis:** The problem provides a clear formula $D = \frac{dv}{60n}$. Students need to identify the relationship between D and n. When d and v remain constant, D is inversely proportional to n: $D \propto \frac{1}{n}$. This means that if n is doubled (multiplied by 2), D is halved (divided by 2). This is a direct application of fundamental algebraic relationships and algorithmic rules, without requiring actual calculation of specific numerical values.

2. Articulate a solution, showing and/or summarising and presenting intermediate mathematical results (Point 8 in the list)

The question asks to "describe precisely" how D changes. This requires students to express the conclusion using clear, accurate mathematical language.

- **Specific Analysis:** Students need to clearly state the conclusion: "The value of D will be halved" or " D is divided by 2". This demonstrates the ability to translate the mathematical reasoning process (identifying the inverse relationship) into a precise, understandable answer.

3. Manipulate numbers, graphical and statistical data and information, algebraic expressions and equations, and geometric representations (Point 7 in the list)

The task requires students to manipulate an algebraic equation.

- **Specific Analysis:** Students need to manipulate the formula $D = \frac{dv}{60n}$ by substituting n with $2n$, resulting in $D_{new} = \frac{dv}{60(2n)} = \frac{1}{2} \times \frac{dv}{60n} = \frac{1}{2} D_{old}$. This process involves manipulating and simplifying algebraic expressions.

4. Use a multi-step procedure leading to a mathematical solution, conclusion or generalisation (Point 12 in the list)

Despite its apparent simplicity, the solution process involves multiple logical steps.

- **Specific Analysis:**
  1. Understand the problem statement and the formula.
  2. Identify the key variables and the variables that remain constant.
  3. Apply algebraic rules to determine the effect of the change in n on D (inverse relationship).
  4. Derive and clearly express the final conclusion.
     This is a multi-step logical process involving analysis, application of rules, and summation.

1. 利用基于定义、规则和形式系统的构建，包括运用熟悉的算法 (列表中的第 5 点)

这是解决此问题的核心能力。学生必须理解并应用数学规则，特别是关于变量之间反比例关系的代数规则。

- **具体分析：** 题目提供了一个明确的公式 $D = \frac{dv}{60n}$。学生需要识别 D 和 n 之间的关系。当 d 和 v 保持不变时，D 与 n 成反比关系：$D \propto \frac{1}{n}$。这意味着如果 n 增加一倍（乘以 2），D 就会减少一半（除以 2）。这是对基础代数关系和算法规则的直接应用，无需实际计算具体数值。

2. 阐明解决方案，展示和/或总结并呈现中间数学结果 (列表中的第 8 点)

题目要求"精确描述"D 如何变化。这要求学生用清晰、准确的数学语言表达结论。

- **具体分析：** 学生需要清晰地表述结论："D 的值将减半"或"D 除以 2"。这体现了将数学推理过程（识别反比关系）转化为准确、可理解的答案的能力。

3. 操作数字、图形和统计数据及信息、代数表达式和方程式以及几何表示 (列表中的第 7 点)

该任务要求学生操作一个代数方程式。

- **具体分析：** 学生需要操作公式 $D = \frac{dv}{60n}$，通过将 n 替换为 $2n$，得到 $D_{new} = \frac{dv}{60(2n)} = \frac{1}{2} \times \frac{dv}{60n} = \frac{1}{2} D_{old}$。这个过程涉及对代数表达式进行操作和简化。

4. 利用多步骤程序得出数学解、结论或概括 (列表中的第 12 点)

尽管看似简单，解决过程涉及多个逻辑步骤。

- **具体分析：**
  1. 理解问题陈述和公式。
  2. 识别关键变量和保持不变的变量。
  3. 应用代数规则确定 n 变化对 D 的影响（反比关系）。
  4. 得出并清晰表达最终结论。
     这是一个涉及分析、应用规则和总结的多步骤逻辑过程。